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Robust control of aerial cable-suspended payload transportation via fully actuated system approach

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ABSTRACT

Unmanned aerial vehicles with cable-suspended payloads suffer from underactuation, strong coupling, and disturbances. These factors often cause payload swing and degrade tracking performance. This paper proposes a robust control method based on the fully actuated system (FAS) framework. The system dynamics are reformulated into a reduced-order FAS form with virtual constraints. The remaining states are arranged into a cascade structure for ensuring stability. On this basis, FAS controllers are designed for both outer-loop dynamics (UAV position and payload swing) and inner-loop attitude dynamics. Disturbance observers are introduced to handle external disturbances. Stability is proved through cascade analysis. Simulations and experiments confirm that the proposed controllers achieve robust waypoint tracking with improved swing suppression.

1. Introduction

In recent years, unmanned aerial vehicles (UAVs) equipped with cable-suspended payloads have emerged as an effective solution for aerial transportation tasks such as disaster relief, precision positioning, and various industrial applications. Compared to rigidly mounted payloads, the slung-payload configuration offers greater flexibility and efficiency, enabling UAVs to transport heavy or unwieldy objects in complex environments (Palunko et al., 2012; Villa et al., 2020). However, this flexible attachment also introduces additional dynamic complexity, where the payload behaves as a pendulum, resulting in coupled oscillations that challenge both flight safety and the control accuracy (Guerrero et al., 2015; Kang et al., 2023).

The inherent difficulty lies in the underactuated nature of the UAV cable-suspended payload systems. A multi-rotor UAV is usually underactuated, as it has six degrees of freedom but only four independent control inputs. Suspending a payload via a cable introduces two additional swing degrees of freedom that are not directly actuated, making the overall system highly underactuated (Kang & Shan, 2025b; Liang et al., 2018). Moreover, this structural underactuation leads to strong nonlinear couplings, compounded by uncertainties in payload mass and tether length, as well as external disturbances such as wind gusts (Akhtar et al., 2023; Qian & Liu, 2020; Zheng et al., 2025). These factors may induce payload oscillations, degrade tracking accuracy, or even destabilize the system during aggressive flight. Consequently, ensuring robust stability and swing suppression remains a critical issue for reliable aerial transportation.

To address these challenges, various control strategies have been explored. Hierarchical cascade designs exploit the separation between inner-loop attitude and outer-loop translational dynamics (Kang et al., 2023; Liang et al., 2018; Lv et al., 2020). Passivity-based methods have been proposed to improve swing damping performance (Guerrero et al., 2015), while adaptive and learning-based controllers have been developed to compensate for uncertain payload parameters (Kang & Shan, 2025a; Liang et al., 2021). In addition, sliding mode control and disturbance-observer techniques are widely adopted in practice to attenuate external disturbances (Chandra & Lal, 2022; Liu et al., 2022; Yang et al., 2024); for instance, an intermediate-observer-based active disturbance rejection control (ADRC) has been used to achieve robust and accurate system performance in the presence of disturbances and delays (Liu et al., 2022). More recently, virtual constraint-based approaches have been investigated to exploit nonlinear couplings for enhanced swing suppression performance (Kang & Shan, 2025b; Kang et al., 2025). Despite these advances, many existing designs depend on accurate system models, require restrictive assumptions, or suffer from practical implementation challenges such as input chattering or sensitivity to initial conditions.

Beyond these conventional designs, the Fully Actuated System (FAS) methodology has emerged as a promising framework for nonlinear and underactuated systems (Duan, 2021, 2024a,b). By converting an underactuated system into an equivalent FAS, the control design can be significantly simplified while improving robustness against uncertainties and disturbances. However, directly applying the conventional FAS procedure to the UAV cable-suspended payload system is not

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straightforward. The main challenges lie in: (i) the strong inertial coupling between translational and swing states, and (ii) the UAV force input distribution in both translational and swing motions. These characteristics make it very difficult to construct the diffeomorphic coordinate transformation in the FAS design (Duan, 2024b).

To address these issues, this paper reformulates the outer-loop dynamics into a structure that is more amenable to the FAS framework. Building on virtual constraints (Kang et al., 2025), a reduced-order FAS representation is derived, while the remaining dynamics are expressed in a cascade form whose stability can be established through cascade theory. In particular, the introduced virtual constraint avoids the need for constructing a full diffeomorphic transformation. By embedding the underactuated swing dynamics into a composite coordinate, it yields a tractable reduced-order FAS subsystem. Furthermore, disturbance observers are incorporated into the FAS-based controller design to enhance robustness against model uncertainties and external disturbances. To the best of our knowledge, this is the first attempt to apply the FAS approach to the UAV cable-suspended payload system. The main contributions of this work are summarized as follows:

(i) A reduced-order FAS formulation is developed for UAV cable-suspended payload systems using virtual constraints. The proposed formulation reorganizes the underactuated outer-loop dynamics into a tractable cascade structure.
(ii) Disturbance observer-based FAS controllers are designed for both the outer-loop and inner-loop subsystems. The proposed controllers enhance robustness against external disturbances.
(iii) Rigorous stability is established via cascade analysis. Simulation and experimental results further demonstrate stable waypoint tracking and effective swing suppression.

The following lemma will be useful for stability analysis in subsequent sections.

Lemma 1 (Shtessel et al. (2007)). *Consider the following differential inclusion for $e_i \in \mathbb{R},\ i = 0, 1, \ldots, n$:*

$$\begin{cases} \dot{e}_0 = -\lambda_0 \rho^{1/(n+1)} \lceil e_0 \rfloor^{n/(n+1)} + e_1 \\ \dot{e}_1 = -\lambda_1 \rho^{1/n} \lceil e_1 - \dot{e}_0 \rfloor^{(n-1)/n} + e_2 \\ \vdots \\ \dot{e}_{n-1} = -\lambda_{n-1} \rho^{1/2} \lceil e_{n-1} - \dot{e}_{n-2} \rfloor^{1/2} + e_n \\ \dot{e}_n \in -\lambda_n \rho \,\mathrm{sign}(e_n - \dot{e}_{n-1}) + [-\rho, \rho]. \end{cases}$$

where the parameters $\lambda_i > 0$ are sufficiently large, $\rho > 0$ is a constant, and $\lceil x \rfloor^{\alpha} := |x|^{\alpha}\,\mathrm{sign}(x)$. Then, the origin $e_i = 0$ for all $i = 0, \ldots, n$ is finite-time stable.

2. Modeling and problem statement

2.1. System dynamics

We consider an aerial transportation system in which a payload is attached to the geometric center of a UAV via a lightweight rigid cable of fixed length L, as illustrated in Fig. 1. The UAV is modeled as a rigid body with mass M, while the payload is considered as a point mass of m. The UAV position $(x, y, z) \in \mathbb{R}^3$ is defined in the inertial frame $\mathcal{I} = \{e_1, e_2, e_3\}$. The payload motion relative to the UAV is parameterized by two swing angles $\{|\theta_{1,2}| < \pi/2\} \subset \mathbb{R}^2$.

The UAV's translational dynamics are described by (1)-(3), together with the cable swing dynamics Eqs. (4) and (5) associated with the cable-suspended payload.

$$(M+m)\ddot{x} + mL\ddot{\theta}_1 c_{\theta_1} c_{\theta_2} - mL\ddot{\theta}_2 s_{\theta_1} s_{\theta_2} - mL(\dot{\theta}_1^2 s_{\theta_1} c_{\theta_2} + 2\dot{\theta}_1\dot{\theta}_2 c_{\theta_1} s_{\theta_2} + \dot{\theta}_2^2 s_{\theta_1} s_{\theta_2}) + d_{ux} = u_x \tag{1}$$

$$(M+m)\ddot{y} + mL\ddot{\theta}_2 c_{\theta_2} - mL\dot{\theta}_2^2 s_{\theta_2} + d_{uy} = u_y \tag{2}$$

$$(M+m)\ddot{z} + mL\ddot{\theta}_1 s_{\theta_1} c_{\theta_2} + mL\ddot{\theta}_2 c_{\theta_1} s_{\theta_2} + mL(\dot{\theta}_1^2 c_{\theta_1} c_{\theta_2} - 2\dot{\theta}_1\dot{\theta}_2 s_{\theta_1} s_{\theta_2} + \dot{\theta}_2^2 c_{\theta_1} c_{\theta_2}) + (M+m)g + d_{uz} = u_z \tag{3}$$

$$mL\ddot{x} c_{\theta_1} c_{\theta_2} + mL\ddot{z} s_{\theta_1} c_{\theta_2} + mL^2\ddot{\theta}_1 c_{\theta_2}^2 - 2mL^2\dot{\theta}_1\dot{\theta}_2 c_{\theta_2} s_{\theta_2} + mLg s_{\theta_1} c_{\theta_2} = 0, \tag{4}$$

$$-mL\ddot{x} s_{\theta_1} s_{\theta_2} + mL\ddot{y} c_{\theta_2} + mL\ddot{z} c_{\theta_1} s_{\theta_2} + mL^2\ddot{\theta}_2 + mL^2\dot{\theta}_1^2 s_{\theta_2} c_{\theta_2} + mLg c_{\theta_1} s_{\theta_2} = 0. \tag{5}$$

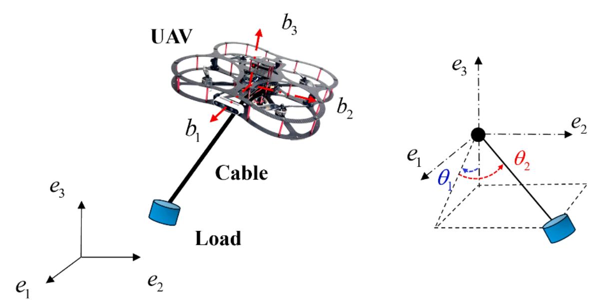

Fig. 1. Configuration of an aerial cable-suspended payload system.

Here $u = [u_x, u_y, u_z]^{\top} \in \mathbb{R}^3$ denotes the translational control force generated by the total thrust F along the body-fixed third axis b_3. Specifically, $u = Fb_3 = FRe_3$, where $R \in \mathrm{SO}(3)$ is the rotation matrix from the body-fixed frame $\mathcal{B} = \{b_1, b_2, b_3\}$ to the inertial frame $\mathcal{I}$, and $e_3 = [0, 0, 1]^{\top}$. The constant g denotes the gravitational acceleration. The vector $d_u = [d_{ux}, d_{uy}, d_{uz}]^{\top}$ represents external disturbances acting on the translational motion. For notational compactness, we use $s_{\theta_i} = \sin(\theta_i)$ and $c_{\theta_i} = \cos(\theta_i)$ for $i \in \{1, 2\}$.

The rotational dynamics of the UAV are described by

$$\dot{R} = R\omega^{\times}, \tag{6}$$

$$J\dot{\omega} + \omega^{\times} J\omega + d_{\tau} = \tau, \tag{7}$$

where $J \in \mathbb{R}^{3\times3}$ is the inertia matrix, $\omega \in \mathbb{R}^3$ is the angular velocity, and the control torque $\tau = (\tau_x, \tau_y, \tau_z) \in \mathbb{R}^3$ directly actuates the UAV attitude. d_{τ} denotes the external disturbances affecting the UAV attitude motion. For any vector $x = [x_1, x_2, x_3]^{\top} \in \mathbb{R}^3$, the skew-symmetric operator $(\cdot)^{\times} : \mathbb{R}^3 \to \mathbb{R}^{3\times3}$ is defined such that $x^{\times} y = x \times y$ for all $y \in \mathbb{R}^3$, i.e.,

$$x^{\times} = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}.$$

2.2. Problem formulation

From (1)–(7), it is clear that the aerial cable-suspended payload system is both underactuated and strongly coupled. The system has eight degrees of freedom, consisting of the six coordinates of the UAV and the two swing angles of the cable, while only four independent inputs $(F, \tau_x, \tau_y, \tau_z)$ are available. This mismatch leaves four degrees of underactuation, which fundamentally characterizes the complexity of the system. Due to the high underactuation degrees, both the UAV translational motions and payload swing dynamics cannot be directly actuated and can only be regulated indirectly through the UAV attitude. In addition, external disturbances and system uncertainties further amplify this coupling effect. As a result, the control design must simultaneously ensure accurate UAV position regulation while effective suppressing the payload oscillations, which reflects the intrinsic difficulty of underactuated control problems. The formal statement of the control problem is given as follows.

Control Problem 1: Consider the aerial cable-suspended payload system (1)–(7) with control inputs $(F, \tau_x, \tau_y, \tau_z)$. The control objective is to achieve the asymptotic stabilization in the presence of bounded disturbances such that: 1) the UAV position reaches the desired location

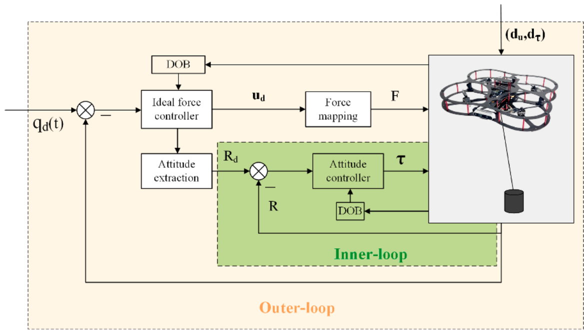

Fig. 2. Hierarchical control framework.

(x_d, y_d, z_d); 2) The payload swing angles converge to zero asymptotically, i.e., $(\theta_1, \theta_2) \to (0, 0)$. Formally, the closed-loop system is required to satisfy $(x(t), y(t), z(t), \theta_1(t), \theta_2(t)) \to (x_d, y_d, z_d, 0, 0)$ as $t \to \infty$.

2.3. Hierarchical control architecture

In this section, we introduce a hierarchical control structure for UAV cable-suspended payload transportation, following the general framework in (Kang et al., 2023). As illustrated in Fig. 2, the control system consists of two loops: an inner loop for UAV attitude control and an outer loop for regulating the UAV position and suppressing payload swing motion. To facilitate the outer-loop design, a virtual force command $u_d = [u_{dx}, u_{dy}, u_{dz}]^\top$ is introduced as $u_d := F R_d e_3$, where R_d denotes the desired attitude. The actual translational force $u = F R e_3$ converges to u_d as $R \to R_d$ under inner-loop attitude tracking. The inner loop is responsible for stabilizing the UAV attitude dynamics, which are directly actuated by the control torque. This allows the UAV to accurately track the desired attitude R_d and generate the commanded force vector u_d. On top of this, the outer loop addresses the underactuated translational dynamics and payload swing suppression by generating the desired force command u_d based on the UAV position, payload oscillation states, and external disturbances.

The hierarchical structure separates the fast attitude dynamics from the slower translational and swing motions, enabling tractable controller design and robust stabilization of the overall underactuated system.

3. Outer-loop control via FAS

3.1. Cascade and FAS formulation

The outer-loop dynamics are defined on the translational and swing coordinates (r, θ), and can be written as the coupled system

$$\ddot{r} = M_a^{-1}(u_d + f), \tag{8}$$

$$M_u(\theta)\,\ddot{\theta} + C_u(\theta, \dot{\theta})\,\dot{\theta} + G_u(\theta) = S(\theta)\ddot{r}, \tag{9}$$

where $r = [x, y, z]^\top$ and $\theta = [\theta_1, \theta_2]^\top$. Eq. (8) represents the translational dynamics of the UAV obtained via partial feedback linearization (Spong, 1994), whereas (9) describes the unactuated payload swing dynamics in the Euler-Lagrange form. It is easy to see that the system exhibits the cascade coupling structure, in which the translational acceleration $\ddot{r}$ acts as a driving input to the swing dynamics.

For clarity in developing the FAS formulation, only the nominal system is considered at this stage, and external disturbances are temporarily neglected. The corresponding system matrices are given below for completeness.

$$M_a = \begin{bmatrix} M + m s_{\theta_1}^2 c_{\theta_2}^2 & m s_{\theta_1} s_{\theta_2} c_{\theta_2} & -m s_{\theta_1} c_{\theta_1} c_{\theta_2}^2 \\ m s_{\theta_1} s_{\theta_2} c_{\theta_2} & M + m s_{\theta_2}^2 & -m c_{\theta_1} s_{\theta_2} c_{\theta_2} \\ -m s_{\theta_1} c_{\theta_1} c_{\theta_2}^2 & -m c_{\theta_1} s_{\theta_2} c_{\theta_2} & M + m c_{\theta_1}^2 c_{\theta_2}^2 \end{bmatrix}.$$

$$M_u(\theta) = \begin{bmatrix} L c_{\theta_2}^2 & 0 \\ 0 & L \end{bmatrix}, \quad G_u(\theta) = \begin{bmatrix} g c_{\theta_2} s_{\theta_1} \\ g c_{\theta_1} s_{\theta_2} \end{bmatrix},$$

$$C_u(\theta, \dot{\theta}) = \begin{bmatrix} 0 & -2 L c_{\theta_2} s_{\theta_2} \dot{\theta}_1 \\ L c_{\theta_2} s_{\theta_2} \dot{\theta}_1 & 0 \end{bmatrix},$$

$$S(\theta) = \begin{bmatrix} -c_{\theta_1} c_{\theta_2} & 0 & -s_{\theta_1} c_{\theta_2} \\ s_{\theta_1} s_{\theta_2} & -c_{\theta_2} & -c_{\theta_1} s_{\theta_2} \end{bmatrix},$$

and

$$f = \begin{bmatrix} m(L\dot{\theta}_1^2 s_{\theta_1} c_{\theta_2}^3 + L\dot{\theta}_2^2 s_{\theta_1} s_{\theta_2} + g c_{\theta_1} s_{\theta_1} c_{\theta_2}^2) \\ m(L\dot{\theta}_1^2 c_{\theta_2}^2 + L\dot{\theta}_2^2 + g c_{\theta_1} c_{\theta_2}) s_{\theta_2} \\ -Mg - m(g c_{\theta_1}^2 c_{\theta_2}^2 + L\dot{\theta}_1^2 c_{\theta_1} c_{\theta_2}^3 + L\dot{\theta}_2^2 c_{\theta_1} c_{\theta_2}) \end{bmatrix}.$$

It should be noted that directly extending the conventional FAS approach to the outer-loop dynamics (8)-(9) is not straightforward. Although the FAS approach has been successfully applied to several classes of underactuated mechanical systems in (Duan, 2024b), the present coupled outer-loop dynamics do not fit into any of these types. The main difficulty lies in the fact that the translational acceleration $\ddot{r}$ not only depends on the control input u_d but also appears explicitly as a forcing term in the unactuated swing dynamics. This structural coupling prevents the direct construction of the diffeomorphic coordinate transformation required by the conventional FAS methodology.

This motivates a reformulation of the outer-loop dynamics into a structure amenable to FAS-based stabilization. Specifically, we consider a representation that partially exhibits the properties of a reduced-order FAS subsystem, while the remaining dynamics form a cascade whose stability can be established via cascade analysis. To this end, we now introduce a new coordinate transformation

$$\begin{cases} \xi = \dot{r} + \varphi(r, \theta) \\ \dot{\xi} = \ddot{r} + \dot{\varphi}(r, \theta), \end{cases} \tag{10}$$

where $\varphi(r, \theta) : \mathbb{R}^3 \times \mathbb{R}^2 \to \mathbb{R}^3$ is a smooth mapping satisfying $\varphi(r_d, 0) = 0$, and its derivative is $\dot{\varphi} = \varphi_r(r, \theta)\dot{r} + \varphi_\theta(r, \theta)\dot{\theta}$ with $\varphi_r(r, \theta) = \frac{\partial \varphi}{\partial r} \in \mathbb{R}^{3\times 3}$ and $\varphi_\theta(r, \theta) = \frac{\partial \varphi}{\partial \theta} \in \mathbb{R}^{3\times 2}$. The variable ξ is introduced as a composite (coupling) signal that links the actuated translational dynamics with the underactuated swing motion through the mapping $\varphi(r, \theta)$. The mapping $\varphi(r, \theta)$ is designed such that the desired equilibrium $(r, \theta, \dot{r}, \dot{\theta}) = (r_d, 0, 0, 0)$ corresponds to $\xi = 0$ (i.e., $\varphi(r_d, 0) = 0$), and the resulting cascade representation enables the swing dynamics to be indirectly regulated via the translational channel.

Under this coordinate transformation, the outer-loop dynamics can be rewritten in a cascade form. The first subsystem is given by

$$\Sigma_1 : \begin{cases} \dot{r} \; = \; -\varphi + \xi, \\ \ddot{r} \; = \; -\dot{\varphi} + \dot{\xi}, \\ \ddot{\theta} \; = \; -M_u^{-1}\big(C_u \dot{\theta} + G_u\big) - M_u^{-1} S(\dot{\varphi} - \dot{\xi}), \end{cases} \tag{11}$$

which describes the coupled translational and swing dynamics driven by the composite signal ξ.

The second subsystem is defined as

$$\Sigma_2 : \quad \dot{\xi} = v_o \tag{12}$$

where $v_o \in \mathbb{R}^3$ is the new input to be designed for the outer-loop system. Accordingly, the actual control input is given as

$$u_d = M_a(v_o - \dot{\varphi}) - f \tag{13}$$

Notably, with an appropriate choice of the mapping φ, the subsystem Σ_1 is asymptotically stable at the desired equilibrium $(r, \theta) = (r_d, 0)$ when $\xi = \dot{\xi} = 0$. In this case, if the auxiliary coordinate ξ vanishes, the translational and swing states will approach the equilibrium configuration. As a result, the outer-loop control problem reduces to stabilizing the FAS subsystem Σ_2, which can be accomplished using standard FAS techniques. Stabilization of Σ_2, together with the cascade interconnection, ensures asymptotic stability of the overall outer-loop dynamics and provides a systematic framework for underactuated control design.

3.2. Disturbance-observer-based controller design

To address external disturbances and model uncertainties, we augment the given subsystem (12) by including an equivalent disturbance, such that

$$\dot{\xi} = v_o + d_o, \tag{14}$$

where $d_o \in \mathbb{R}^3$ denotes a lumped matched disturbance. In particular, the translational dynamics of the UAV are perturbed by an additive disturbance, the translational dynamics can be rewritten as $\ddot{r} = M_a^{-1}(u+f) + d_o$ from (1)-(3), see (Kang et al., 2025).

Considering (14), the input v_o is designed within the FAS framework as

$$v_o = -K_2\,\xi - \hat{d}_o, \tag{15}$$

where $K_2 \in \mathbb{R}^{3\times 3}$ is a positive definite gain matrix, and $\hat{d}_o$ is the disturbance estimate provided by the following disturbance observer,

$$\begin{cases} \dot{z}_\xi = -\lambda_1 \rho_o^{1/2} \lceil z_\xi - \xi \rfloor^{1/2} + v_o + \hat{d}_o, \\ \dot{\hat{d}}_o = -\lambda_2 \rho_o \operatorname{sign}(z_\xi - \xi), \end{cases} \tag{16}$$

where $\rho_o > 0$ is a design constant satisfying $\sup_{t\geq 0} \|\dot{d}_o(t)\| < \rho_o$, and the observer gains $\lambda_{1,2}$ are positive constants.

As a result, the closed-loop dynamics of the FAS subsystem reduce to

$$\dot{\xi} = -K_2\,\xi - \tilde{d}_o, \tag{17}$$

where $\tilde{d}_o = \hat{d}_o - d_o$ denotes the error of the estimated disturbance. In the nominal case $\tilde{d}_o = 0$, it is easy to obtain that the subsystem (17) is exponentially stable at the origin.

Once the disturbance observer-based controller ensures effective stabilization of (17), the remaining step is to construct a suitable mapping $\varphi(r,\theta)$ such that the subsystem (11) is stable. To this end, motivated by our earlier work on virtual constraints (Kang & Shan, 2025b; Kang et al., 2025), we consider

$$\varphi(r,\theta) := K_1\big(r - r_d + \pi_\theta\big), \tag{18}$$

where $K_1 \in \mathbb{R}^{3\times 3}$ is positive definite, and

$$\pi_\theta := \delta \begin{bmatrix} -s_{\theta_1} c_{\theta_2} \\ -s_{\theta_2} \\ c_{\theta_1} c_{\theta_2} - 1 \end{bmatrix}$$

where $\delta > 0$ is a design parameter and π_θ vanishes at $\theta = 0$, thereby ensuring the equilibrium at $(r,\theta) = (r_d, 0)$.

Consequently, substituting (18) into (10) yields the auxiliary coordinate as

$$\xi = \dot{r} + K_1(r - r_d + \pi_\theta). \tag{19}$$

Thus, the outer-loop controller (13) becomes

$$u_d = -M_a\big(K_2\,\xi + \hat{d}_o + \dot{\varphi}\big) - f, \tag{20}$$

where $\dot{\varphi} = K_1\,(\dot{r} + \dot{\pi}_\theta)$ with

$$\dot{\pi}_\theta = \delta \begin{bmatrix} -c_{\theta_1} c_{\theta_2}\,\dot{\theta}_1 + s_{\theta_1} s_{\theta_2}\,\dot{\theta}_2 \\ -c_{\theta_2}\,\dot{\theta}_2 \\ -s_{\theta_1} c_{\theta_2}\,\dot{\theta}_1 - c_{\theta_1} s_{\theta_2}\,\dot{\theta}_2 \end{bmatrix} = \delta S(\theta)^\top \dot{\theta}.$$

3.3. Stability analysis

To analyze the stability of the closed-loop outer-loop dynamics, we divide the analysis into two parts. First, we study the FAS subsystem associated with the coordinates ξ, for which the DOB-based controller ensures disturbance rejection and stabilization. Second, we investigate the stability of the translational-swing subsystem under $\xi = 0$. By combining these results through a cascade argument, we will establish that the outer-loop dynamics under the FAS controller (20) is asymptotically stable.

Theorem 1. *Consider the cascade system composed of Σ_1 in (11) and Σ_2 in (14) under the control law (15). The closed-loop FAS subsystem in (17) is exponentially stable at the origin. Moreover, the subsystem Σ_1 is asymptotically stable at $(r,\theta) = (r_d, 0)$.*

Proof. We first show that the DOB system (16) is finite-time stable.

Let $e_o = z_\xi - \xi$ and $\tilde{d}_o = \hat{d}_o - d_o$ denote the observer errors. Then, from (14) and (16), we can obtain the following differential inclusion describing the error dynamics

$$\begin{cases} \dot{e}_o = -\lambda_1 \rho_o^{1/2} \lceil e_o \rfloor^{1/2} + \tilde{d}_o, \\ \dot{\tilde{d}}_o \in -\lambda_2 \rho_o \operatorname{sign}(\tilde{d}_o - \dot{e}_o) \; + \; [-\rho_o, \rho_o]. \end{cases} \tag{21}$$

By applying Lemma 1, the differential inclusion (21) is finite-time stable at the origin. Therefore, there exists a finite time $t_1 > 0$ such that $e_o = 0$ and $\tilde{d}_o = 0$ for all $t \geq t_1$. Moreover, we can have $e_o \in \mathcal{L}_\infty$ and $\tilde{d}_o \in \mathcal{L}_\infty$. Hence, e_o and $\tilde{d}_o$ are bounded for all $t \geq 0$.

Then, we establish the asymptotic stability of the FAS subsystem (17) by the following Lyapunov function

$$V_1 = \frac{1}{2}\xi^\top \xi. \tag{22}$$

Taking the time derivative of V_1 along the (17) yields that

$$\dot{V}_1 = -\,\xi^\top K_2 \xi - \xi^\top \tilde{d}_o. \tag{23}$$

Since the estimation error $\tilde{d}_o$ vanishes after a finite time, i.e., $\tilde{d}_o = 0$ for all $t \geq t_1$, (23) becomes

$$\dot{V}_1 \leq -\lambda_{\min}(K_2)\xi^\top \xi \leq -2\lambda_{\min}(K_2) V_1, \tag{24}$$

where $\lambda_{\min}(K_2)$ denotes the minimum eigenvalue of K_2. This implies the exponential stability of closed-loop FAS subsystem for $t \geq t_1$.

Moreover, in the transient interval $[0, t_1)$, from (23) one has

$$\begin{aligned} \dot{V}_1 &\leq -\lambda_{\min}(K_2)\,\xi^\top \xi + \tfrac{\lambda_{\min}(K_2)}{2}\,\xi^\top \xi + \tfrac{1}{2\lambda_{\min}(K_2)}\,\|\tilde{d}_o\|^2 \\ &\leq \tfrac{1}{2\lambda_{\min}(K_2)}\,\|\tilde{d}_o\|^2 \end{aligned} \tag{25}$$

Thus, V_1 remains bounded for $t \in [0, t_1)$ because $\tilde{d}_o \in \mathcal{L}_\infty$, implying that ξ is bounded on this interval. Combining the boundedness on $[0, t_1)$ and the exponential stability for $t \geq t_1$, it is concluded that $(\xi, \dot{\xi}) \in \mathcal{L}_\infty$ and $(\xi, \dot{\xi}) \to (0,0)$ as $t \to \infty$.

Finally, we prove the asymptotic stability of the subsystem (11) at $(\xi, \dot{\xi}) = (0,0)$.

Consider a Lyapunov function V_2 as

$$V_2 = \frac{\delta}{2}\dot{\theta}^\top M_u \dot{\theta} + \delta g(1 - c_{\theta_1} c_{\theta_2}) + \frac{1}{2}\dot{r}^\top \dot{r}. \tag{26}$$

Clearly, $V_2 \geq 0$ since M_u is positive definite. Then, taking its derivative along (11) gives

$$\begin{aligned} \dot{V}_2 &= -\delta \dot{\theta}^\top S\,\dot{\varphi} - \dot{r}^\top \dot{\varphi} \\ &= -\dot{\varphi}^\top (\dot{r} + \delta S^\top \dot{\theta}). \end{aligned} \tag{27}$$

Recalling that $\dot{\varphi} = K_1(\dot{r} + \dot{\pi}_\theta)$ and $\dot{\pi}_\theta = \delta S^\top \dot{\theta}$, from (27), one has

$$\dot{V}_2 = -(\dot{r} + \dot{\pi}_\theta)^\top K_1 (\dot{r} + \dot{\pi}_\theta). \tag{28}$$

Thus, $\dot{V}_2 \leq 0$, and $\dot{V}_2 = 0$ only if $\dot{r} + \dot{\pi}_\theta = 0$. Recalling that $\dot{\xi} = \ddot{r} + K_1(\dot{r} + \dot{\pi}_\theta) = 0$, one has $\ddot{r} = 0$. From $\xi \in \mathcal{L}_\infty$ and $\pi_\theta \in \mathcal{L}_\infty$, we have $r \in \mathcal{L}_\infty$. Combining with $\ddot{r} = 0$, one has $\dot{r} = 0$. Consequentially, similar to the analysis in (Kang et al., 2025), it can be obtained that $\pi_\theta = 0$. This indicates that the equilibrium point $(r,\theta) = (r_d, 0)$ is asymptotically reached. Therefore, we can conclude that the outer-loop system will finally asymptotically converge to the desired equilibrium.

This completes the proof. □

4. Inner-loop attitude control via FAS

4.1. Euler-angle parameterization and error dynamics

First, we consider that the UAV attitude is parameterized by a set of Euler angles $\eta = [\phi, \vartheta, \psi]^\top$, where ϕ, ϑ, and ψ denote the roll, pitch, and yaw angles, respectively. For the 3-2-1 (yaw-pitch-roll) rotation sequence, the corresponding rotation matrix is expressed as

$$R = \begin{bmatrix} c_\psi c_\vartheta & c_\psi s_\vartheta s_\phi - s_\psi c_\phi & c_\psi s_\vartheta c_\phi + s_\psi s_\phi \\ s_\psi c_\vartheta & s_\psi s_\vartheta s_\phi + c_\psi c_\phi & s_\psi s_\vartheta c_\phi - c_\psi s_\phi \\ -s_\vartheta & c_\vartheta s_\phi & c_\vartheta c_\phi \end{bmatrix}, \tag{29}$$

and the relation between the Euler angle rates and the body angular velocity is given by

$$\omega = \Pi(\eta)\,\dot\eta, \qquad \Pi(\eta) = \begin{bmatrix} 1 & 0 & -s_\vartheta \\ 0 & c_\phi & s_\phi c_\vartheta \\ 0 & -s_\phi & c_\phi c_\vartheta \end{bmatrix}, \tag{30}$$

where the shorthand notations $c_* = \cos(*)$ and $s_* = \sin(*)$ are used for brevity.

Noting that the quadrotor thrust is always generated along the body-fixed third axis. Hence, the translational (outer-loop) virtual control force $u_d \in \mathbb{R}^3$ can be realized by aligning the thrust direction with u_d through the attitude R_d following (Kang et al., 2023), such that

$$F\, R_d e_3 = u_d, \qquad F = \|u_d\| > 0, \tag{31}$$

which implies $R_d e_3 = u_d/F$ and

$$\begin{cases} u_{dx}/F = c_{\psi_d} s_{\vartheta_d} c_{\phi_d} + s_{\psi_d} s_{\phi_d}, \\ u_{dy}/F = s_{\psi_d} s_{\vartheta_d} c_{\phi_d} - c_{\psi_d} s_{\phi_d}, \\ u_{dz}/F = c_{\vartheta_d} c_{\phi_d}. \end{cases} \tag{32}$$

After some straightforward trigonometric manipulations of (32), we can have the following attitude extraction mapping

$$\begin{cases} \phi_d = \mathrm{asin}[(u_{dx} s_{\psi_d} - u_{dy} c_{\psi_d})/F] \\ \vartheta_d = \mathrm{atan2}[(u_{dx} c_{\psi_d} + u_{dy} s_{\psi_d}), u_{dz}] \\ t \mapsto \psi_d \in C^2 \end{cases} \tag{33}$$

where the yaw reference ψ_d is assigned by high-level requirements. Without loss of generality, we set $\psi_d = 0$ in this study.

Remark 1. The mapping (33) is a widely used attitude extraction for quadrotor UAVs (Lv et al., 2020). The $\mathrm{asin}(\cdot)$ function is always well-defined and bounded in $[-1, 1]$ since $\left|u_{dx}\sin\psi_d - u_{dy}\cos\psi_d\right| \le \sqrt{u_{dx}^2 + u_{dy}^2} \le \|u_d\| = F > 0$. In practical implementation, the desired force should satisfy $u_{dz} > 0$ to ensure non-flip operation and avoid Euler-angle singularities, equivalently $|\vartheta_d| < \pi/2$.

Then, defining attitude tracking errors as

$$\tilde\eta = \eta - \eta_d, \qquad \dot{\tilde\eta} = \dot\eta - \dot\eta_d, \tag{34}$$

the Euler-angle error dynamics are obtained by substituting (30) into the rotational dynamics (7) and expressing the result in Euler coordinates. After straightforward calculations, we obtain

$$\ddot{\tilde\eta} = (J\Pi)^{-1}\Big[-J\,\dot\Pi\,\dot\eta - \big(\Pi\dot\eta\big)^{\times} J\,\big(\Pi\dot\eta\big) + \tau\Big] - \ddot\eta_d + d_{in}, \tag{35}$$

where $d_{in} \in \mathbb{R}^3$ denotes a lumped, matched disturbance capturing unmodeled dynamics and external perturbations. Here, $\dot\Pi(\eta, \dot\eta) = \frac{\partial \Pi}{\partial \phi}\dot\phi + \frac{\partial \Pi}{\partial \vartheta}\dot\vartheta$. We assume $\Pi(\eta)$ is nonsingular, i.e., $|\cos\vartheta| > 0$ (equivalently $|\vartheta| < \pi/2$), which is consistent with the practical non-flip operation.

Note that (35) yields a standard second-order error system in $\tilde\eta$, which enables a straightforward attitude-loop design (e.g., pole placement) and convenient incorporation of disturbance rejection terms. This second-order error form is also consistent with the FAS approach (Duan, 2024a), which facilitates systematic controller design and disturbance attenuation.

4.2. Disturbance-observer-based inner-loop controller

Building upon the FAS reformulation, the attitude error dynamics (35) can be compactly expressed as

$$\ddot{\tilde\eta} = v_{in} + d_{in}, \tag{36}$$

where $v_{in} \in \mathbb{R}^3$ is an auxiliary inner-loop input to be designed. Accordingly, the actual control torque τ is obtained through the following input transformation:

$$\tau = J\dot\Pi\dot\eta + \big(\Pi\dot\eta\big)^{\times}\big(J\Pi\dot\eta\big) + J\Pi\big(v_{in} + \ddot\eta_d\big). \tag{37}$$

Based on (36), the control objective reduces to designing the auxiliary input v_{in} such that $\tilde\eta \to 0$ in the presence of the lumped matched disturbance d_{in}. To attenuate the disturbance effect, we incorporate a continuous finite-time disturbance observer to estimate d_{in} online and compensate it in the inner-loop control law.

To this end, define the disturbance estimate $\hat d_{in}$ and an auxiliary observer state z_η as

$$\begin{cases} \dot z_\eta = -\lambda_3 \rho_\eta^{1/2} \lceil z_\eta - \dot{\tilde\eta} \rfloor^{1/2} + v_{in} + \hat d_{in}, \\ \dot{\hat d}_{in} = -\lambda_4 \rho_\eta\, \mathrm{sign}(z_\eta - \dot{\tilde\eta}), \end{cases} \tag{38}$$

where $\rho_\eta > 0$ is a design constant satisfying $\sup_{t\ge 0}\|\dot d_{in}(t)\| < \rho_\eta$, and $\lambda_{3,4} > 0$ are observer gains. Let $\tilde d_{in} = \hat d_{in} - d_{in}$ be the estimation error. With appropriate gain selection, the observer provides finite-time (practical) convergence of $\tilde d_{in}$, enabling effective disturbance compensation.

The auxiliary input v_{in} is then designed in the FAS framework as a second-order pole-placement law

$$v_{in} = -K_3\,\tilde\eta - K_4\,\dot{\tilde\eta} - \hat d_{in}, \tag{39}$$

where $K_3, K_4 \in \mathbb{R}^{3\times 3}$ are positive definite gain matrices chosen to achieve the desired closed-loop response.

Substituting (39) into (36) yields

$$\ddot{\tilde\eta} + K_4\,\dot{\tilde\eta} + K_3\,\tilde\eta = -\tilde d_{in}. \tag{40}$$

It follows directly from (40) that, in the disturbance-free case ($\tilde d_{in} \equiv 0$), the error dynamics are exponentially stable, with a convergence rate determined by the gain matrices K_3 and K_4. In the presence of bounded $\tilde d_{in}$, the tracking error $\tilde\eta$ remains ultimately bounded, and the bound decreases as the accuracy of the disturbance observer improves.

Remark 2. For the cascaded inner-outer loop structure, the effectiveness of the overall controller relies on a time-scale separation. In particular, the inner-loop attitude dynamics should converge sufficiently fast compared with the outer-loop dynamics. This requires selecting relatively large feedback gains in the attitude controller to guarantee rapid error convergence, thereby allowing the outer loop to operate on a slower time scale with the consideration of a well-stabilized attitude subsystem.

4.3. Stability analysis

We now analyze the closed-loop system stability under the proposed DOB-based inner-loop controller. The objective is to show that the attitude tracking error $\tilde\eta$ converges to the origin in the presence of matched disturbances, while the disturbance observer drives the estimation error $\tilde d_{in}$ to zero in finite time. The main result is summarized as follows.

Theorem 2. *Consider the closed-loop system (40) under the controller (39) and the DOB (38). Assume that $\|\dot d_{in}(t)\| \le \rho_\eta$ for all $t \ge 0$ and choose $K_3, K_4 \in \mathbb{R}^{3\times 3}$ to be positive definite. Then, the estimation error $\tilde d_{in}$ converges to zero in finite time, and the attitude tracking error $\tilde\eta$ converges asymptotically to the origin.*

Proof. First, we show that the disturbance observer system (38) is finite-time stable. Denote $e_{in} = z_\eta - \dot{\tilde\eta}$ and $\tilde d_{in} = \hat d_{in} - d_{in}$. Then, the estimation error dynamics can be written as the following differential inclusion:

$$\begin{cases} \dot e_{in} = -\lambda_3 \rho_\eta^{1/2} \lceil e_{in} \rfloor^{1/2} + \tilde d_{in}, \\ \dot{\tilde d}_{in} \in -\lambda_4 \rho_\eta\, \mathrm{sign}(\tilde d_{in} - \dot e_{in}) \;+\; [-\rho_\eta, \rho_\eta]. \end{cases} \tag{41}$$

As shown in Lemma 1, the differential inclusion (41) is finite-time stable at the origin. Therefore, there exists a finite time $t_2 > 0$ such

Table 1
System parameters.

Parameter	Value	Unit
M	1.2	kg
m	0.2	kg
L	0.5	m
J	diag(0.01, 0.008, 0.014)	$\mathrm{kg \cdot m^2}$
g	9.81	$\mathrm{m/s^2}$

that $e_{in} = 0$ and $\tilde{d}_{in} = 0$ for all $t \geq t_2$. Moreover, both e_{in} and $\tilde{d}_{in}$ remain bounded, i.e., $e_{in} \in \mathcal{L}_\infty$ and $\tilde{d}_{in} \in \mathcal{L}_\infty$.

Next, to establish the asymptotic stability of the closed-loop system (40), we consider the following Lyapunov function

$$V_3 = \frac{1}{2}\dot{\tilde{\eta}}^\top \dot{\tilde{\eta}} + \frac{1}{2}\tilde{\eta}^\top K_3 \tilde{\eta}, \tag{42}$$

with $K_3 > 0$. It is easy to verify that $V_3 \geq 0$ and $V_3 = 0$ if $\dot{\tilde{\eta}} = \tilde{\eta} = 0$.

Taking the time derivative of V_3 gives

$$\begin{aligned}\dot{V}_3 =& \dot{\tilde{\eta}}^\top \ddot{\tilde{\eta}} + \dot{\tilde{\eta}}^\top K_3 \tilde{\eta} \\ =& -\dot{\tilde{\eta}}^\top K_4 \dot{\tilde{\eta}} - \dot{\tilde{\eta}}^\top \tilde{d}_{in}.\end{aligned} \tag{43}$$

where the error dynamics (40) has been used in the last line of (43).

Recalling the fact that the estimation error vanishes after a finite time, i.e., $\tilde{d}_{in} = 0$ for all $t \geq t_2$. In this case, $\dot{V}_3 \leq -\lambda_{\min}(K_4)\|\dot{\tilde{\eta}}\|^2$, which implies the asymptotic stability of the closed-loop system for $t \geq t_2$ based on LaSalle's invariance principle (Khalil, 2002).

During the transient interval $[0, t_2)$, applying Young's inequality to (43) yields

$$\begin{aligned}\dot{V}_3 \leq& -\lambda_{\min}(K_4)\,\|\dot{\tilde{\eta}}\|^2 + \frac{\lambda_{\min}(K_4)\|\dot{\tilde{\eta}}\|^2}{2} + \frac{\|\tilde{d}_{in}\|^2}{2\lambda_{\min}(K_4)} \\ \leq& \frac{\|\tilde{d}_{in}\|^2}{2\lambda_{\min}(K_4)}\end{aligned} \tag{44}$$

Since $\tilde{d}_{in} \in \mathcal{L}_\infty$ on $[0, t_2)$, it follows that $\dot{V}_3$ is bounded from above by a finite constant on this interval. Therefore, $V_3(t)$ remains bounded for all $t \in [0, t_2)$, which implies that both $\tilde{\eta}(t)$ and $\dot{\tilde{\eta}}(t)$ are bounded on $[0, t_2)$. For $t \geq t_2$, the disturbance observer ensures $\tilde{d}_{in}(t) = 0$, and the error dynamics reduce to the disturbance-free exponentially stable system, implying $(\tilde{\eta}(t), \dot{\tilde{\eta}}(t)) \to (0, 0)$ as $t \to \infty$. Combining the boundedness on $[0, t_2)$ and the convergence for $t \geq t_2$, we conclude that the closed-loop signals remain bounded for all $t \geq 0$ and the attitude tracking error satisfies $(\tilde{\eta}, \dot{\tilde{\eta}}) \to (0, 0)$ as $t \to \infty$.

This completes the proof. □

5. Simulation and experimental results

To illustrate the effectiveness of the proposed FAS controllers for both the inner and outer loop systems, numerical simulations are performed on a UAV cable-suspended payload system. The physical parameters of the system are provided in Table 1.

5.1. Waypoint tracking control

In the first simulation study, a waypoint-tracking task is considered to demonstrate the capability of the proposed FAS-based controller in payload transportation. The UAV is required to sequentially pass through several preassigned waypoints while carrying the suspended payload. For a clear comparison of the control performance, the nominal outer-loop system without external disturbances ($d_o = \hat{d}_o = 0$) is considered in this case. The proposed controller (13) is evaluated against a baseline PD scheme, where a standard FAS approach is directly applied to the partial feedback linearized system (8) without the coordinate transformation (10), i.e., by setting $\delta = 0$. Control gains for (13) are set as $K_1 = \mathrm{diag}(1, 1, 1)$, $K_2 = \mathrm{diag}(2, 2, 2)$, and $\delta = 0.5$. For the baseline controller, the same gain matrices are used to ensure a fair comparison.

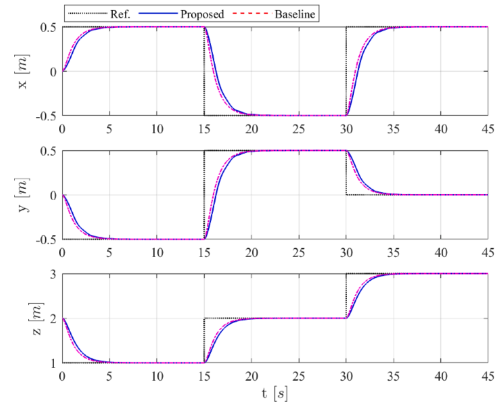

Fig. 3. UAV positions in waypoint tracking. Black dot: reference; Solid blue: proposed controller (FAS (13)); Dashed red: baseline controller (PD, (13) with $\delta = 0$). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

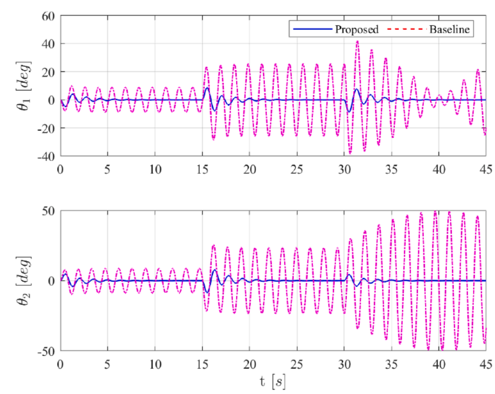

Fig. 4. Swing angles in waypoint tracking.

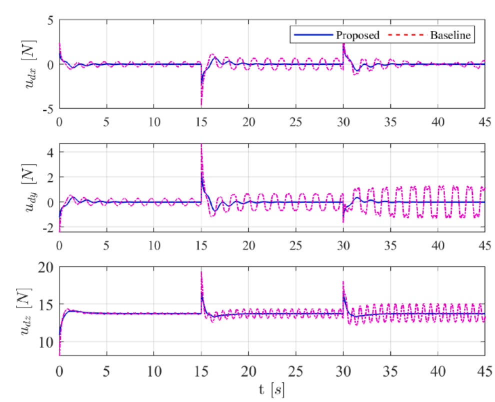

Fig. 5. Desired forces in waypoint tracking.

The initial condition of the system is specified as $(0, 0, 2, 0, 0)$. Three waypoints are assigned for the UAV path, namely $W_1 = (0.5, -0.5, 3)$, $W_2 = (-0.5, 0.5, 2)$, and $W_3 = (0.5, 0, 3)$ (unit: meters).

The simulation results are illustrated in Figs. 3–5. The UAV position tracking performance is depicted in Fig. 3, indicating that both controllers successfully follow the desired waypoints with close convergence and without overshoot. However, as shown in Fig. 4, the proposed controller achieves effective suppression of the payload swing, whereas the baseline controller leads to large oscillations in both swing angles θ_1 and θ_2 during the waypoint transitions. The corresponding control inputs are presented in Fig. 5, where both controllers generate bounded inputs during the entire maneuver. However, the proposed scheme

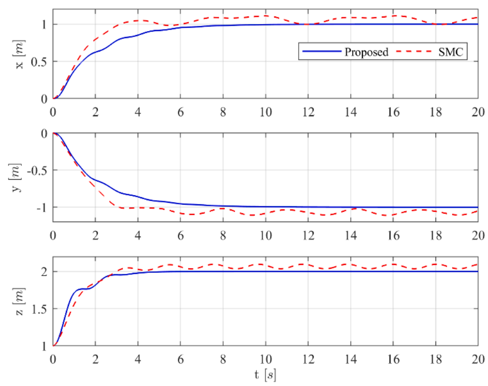

Fig. 6. UAV positions under robust controllers. Solid blue: proposed FAS controller; Dashed red: SMC with smooth $\tanh(\cdot)$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

achieves smoother profiles with reduced oscillations after the waypoint transitions, while the baseline controller produces oscillatory signals, especially in the horizontal forces u_{dx} and u_{dy}. These results indicate that the proposed FAS controller by introducing the virtual constraints not only suppresses the payload swing but also improves the smoothness of the control effort.

5.2. Robust control with disturbances

In the second simulation study, the complete UAV cable-suspended payload system dynamics are considered, including both the outer-loop position control and inner-loop attitude controller. To further assess the robustness of the proposed observer-based FAS control scheme, composite disturbances are applied to the UAV dynamics to emulate wind gusts and model uncertainties.

Specifically, the disturbances acting on the translational (outer-loop) dynamics are defined as

$$d_o(t) = \begin{bmatrix} 0.2\sin(t) + 0.2\cos(2t) + 0.2 \\ 0.3\sin(2t) - 0.1\cos(3t) - 0.2 \\ 0.4\sin(3t) - 0.1\cos(3t) + 0.2 \end{bmatrix}, \quad (45)$$

while those applying to the inner-loop dynamics are chosen as

$$d_\tau(t) = \begin{bmatrix} 0.2\sin(t) + 0.2 \\ -0.2\cos(2t) - 0.2 \\ -0.2\cos(3t) + 0.2 \end{bmatrix}. \quad (46)$$

A conventional sliding-mode controller (SMC) following (Chandra & Lal, 2022) is implemented for comparison. The parameters of the SMC are tuned so as to achieve comparable tracking performance with the proposed controller. For the outer-loop position controller, the control gains are kept the same as before. For the proposed inner-loop attitude controller, the feedback gain matrices are chosen as $K_3 = \text{diag}(20, 20, 20)$, $K_4 = \text{diag}(40, 40, 40)$. The gains for disturbance observers are selected as $\lambda_1 = \lambda_3 = 1.1$ and $\lambda_2 = \lambda_4 = 1.5$, while $\rho_o = \rho_\eta = 10$.

The simulation results under the composite disturbances are illustrated in Figs. 6–9. As shown in Fig. 6, both the proposed FAS controller and the SMC baseline achieve good position tracking; however, the proposed controller yields significantly reduced steady-state tracking errors in all xyz-positions. The payload swing angles shown in Fig. 7 indicates that the proposed controller suppresses oscillations more effectively, with faster decay and smaller residual swings. In contrast, the SMC response exhibits noticeable oscillations because the discontinuous sign function is replaced by the smooth $\tanh(\cdot)$ function, which reduces chattering but weakens disturbance rejection. Similarly, the attitude tracking errors in Fig. 8 confirm that both controllers maintain

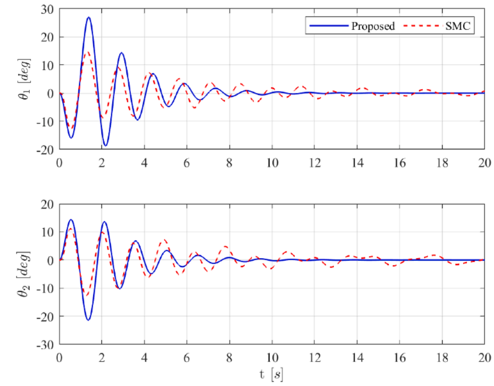

Fig. 7. Swing angles under robust controllers.

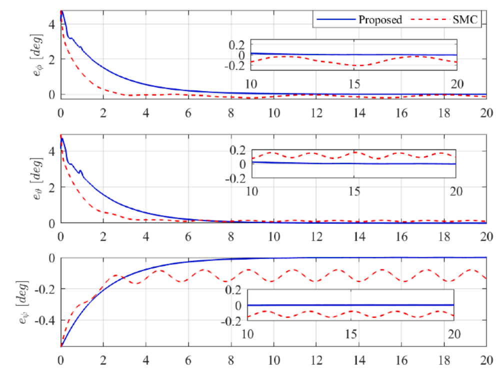

Fig. 8. UAV attitude errors under robust controllers.

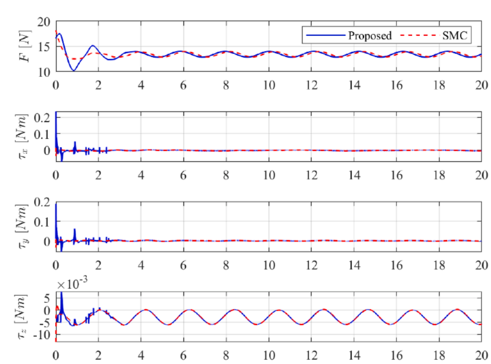

Fig. 9. Control inputs under robust controllers.

bounded tracking performance in the presence of disturbances, but the proposed controller achieves asymptotic convergence. Furthermore, the control inputs depicted in Fig. 9 remain bounded and continuous for both cases. Noting that, for the proposed controller, larger oscillations can be observed in the torque inputs during the initial phase, which is mainly caused by the rapid online update of the disturbance observers. Once the disturbance estimation has converged, the control inputs become smooth after roughly 2 s. Overall, these results demonstrate that introducing virtual constraints through the FAS framework enhances robustness and improves disturbance rejection, particularly in terms of payload swing suppression, when compared with a conventional sliding-mode design.

5.3. Experimental results

In this section, experiments are conducted to validate the proposed FAS-based outer-loop controller and to compare it with the PD and SMC

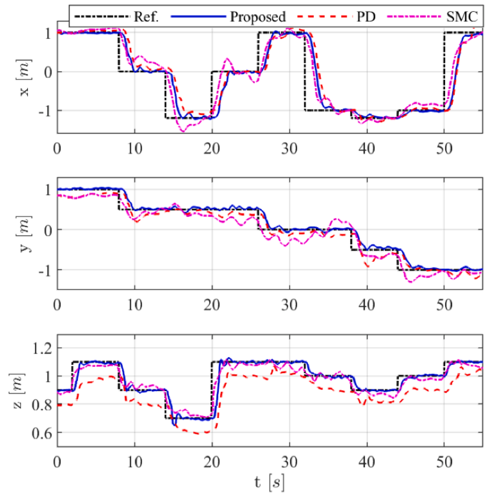

Fig. 10. Experimental UAV positions.

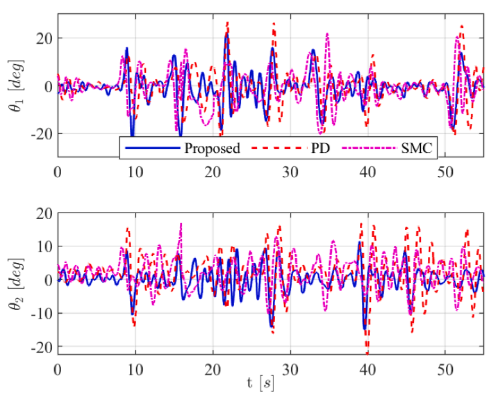

Fig. 11. Experimental payload swing angles.

baselines. The tests are performed on a Quanser QDrone platform inside a motion-capture indoor flying arena. A payload is suspended from QDrone by a lightweight cable, and the UAV position, attitude, and payload motion are measured using an OptiTrack system. The physical parameters of the UAV cable-suspended payload system used in the experiment are consistent with those listed in Table 1. The hardware and software configurations follow the setup described in our previous work (Kang & Shan, 2025a). In the experimental implementation, the attitude dynamics are stabilized by the default onboard controller, and the proposed method is applied at the outer-loop level. For a fair comparison, all controllers are implemented in the same manner, operating under identical inner-loop regulation. All controller parameters are kept identical to the settings in simulations.

In all experimental flights, the UAV is designed to fly through the waypoints in presence of wind disturbances applied along the y-axis. A video of the experimental tests is available at https://youtu.be/9rLxj-GQXXc. The experimental results show noticeable differences among the tested controllers. As shown in Fig. 10, the tracking errors under the proposed method remain small during all transitions, whereas the PD controller shows larger steady tracking offsets, particularly in the y and z directions. The SMC controller achieves smaller offsets but produces larger oscillations. The payload swing angles in Fig. 11 further indicate that the proposed method produces reduced swing amplitudes. The 3D trajectory in Fig. 12 shows that the proposed controller maintains closer tracking of the YU-shaped waypoint layout, while the comparative controllers produce slightly larger deviations. Overall, the experimental results indicate that the observer-based FAS controller

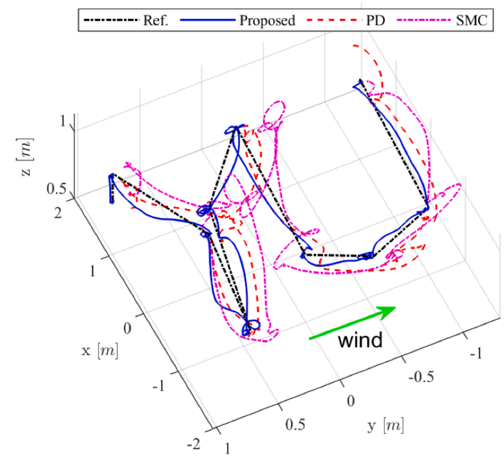

Fig. 12. 3D UAV trajectory: YU-shaped configuration. Solid blue: proposed FAS controller; Dashed red: PD; Dash-dotted magenta: SMC. Fan-generated wind is applied along the y-axis. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

provides consistent tracking and swing suppression performance under practical wind disturbances.

6. Conclusion

This paper studied robust control of UAV cable-suspended payload systems using the FAS approach. By introducing virtual constraints, the coupled dynamics were directly reformulated into a tractable reduced-order FAS error system with a cascade structure. Unlike conventional FAS procedures that require constructing a full diffeomorphic transformation, the proposed approach avoids this cumbersome step and leads to a practically implementable controller/observer design for UAV cable-suspended payload transportation. Closed-loop stability was established via cascade analysis. Simulation and experimental results demonstrated satisfactory position tracking and effective swing suppression, and further showed improved robustness compared with two baseline controllers. Future work will consider extensions to higher-order underactuated systems, such as double-pendulum cable-suspended payload systems.

CRediT authorship contribution statement

Junjie Kang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization; **Jinjun Shan:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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